```
% Clear workspace and figures
clear; close all; clc;

% 1. Define Filter Coefficients (e.g., 4th-order binomial FIR filter)
b = [1, 4, 6, 4, 1] / 16;
a = 1; % FIR filter denominator

% 2. Impulse Responses
N = 128;
[h_direct, n] = impz(b, a, N);

% Cascade Realization using Second-Order Sections (SOS)
[sos, g] = tf2sos(b, a);
delta = [1; zeros(N-1, 1)];
h_cascade = sosfilt(sos, delta) * g;

% --- Figure 1 & 2: Direct and Cascade FIR Impulse Responses ---
figure('Name', 'FIR Impulse Responses', 'Position', [100, 100, 800, 600]);

% Zoomed-in view (matching the first image perspective)[cite: 1]
subplot(2,1,1);
stem(n, h_direct, 'b-o', 'LineWidth', 1.5, 'MarkerFaceColor', 'b');
hold on;
stem(n, h_cascade, 'r--o', 'LineWidth', 1.2, 'MarkerFaceColor', 'none');
title('Direct and Cascade FIR Impulse Responses');
xlabel('Sample n');
ylabel('Amplitude');
legend('Direct Realization', 'Cascade Realization', 'Location', 'northeast');
grid on;
xlim([-2, 10]);

% Full range view (matching the second image perspective)[cite: 2]
subplot(2,1,2);
stem(n, h_direct, 'b-o', 'LineWidth', 1.5, 'MarkerFaceColor', 'b');
hold on;
stem(n, h_cascade, 'r--o', 'LineWidth', 1.2, 'MarkerFaceColor', 'none');
xlabel('Sample n');
ylabel('Amplitude');
legend('Direct Realization', 'Cascade Realization', 'Location', 'northeast');
grid on;
xlim([-2, 130]);

% Frequency Response Calculations
[H, w] = freqz(b, a, 1024);
norm_freq = w / pi;

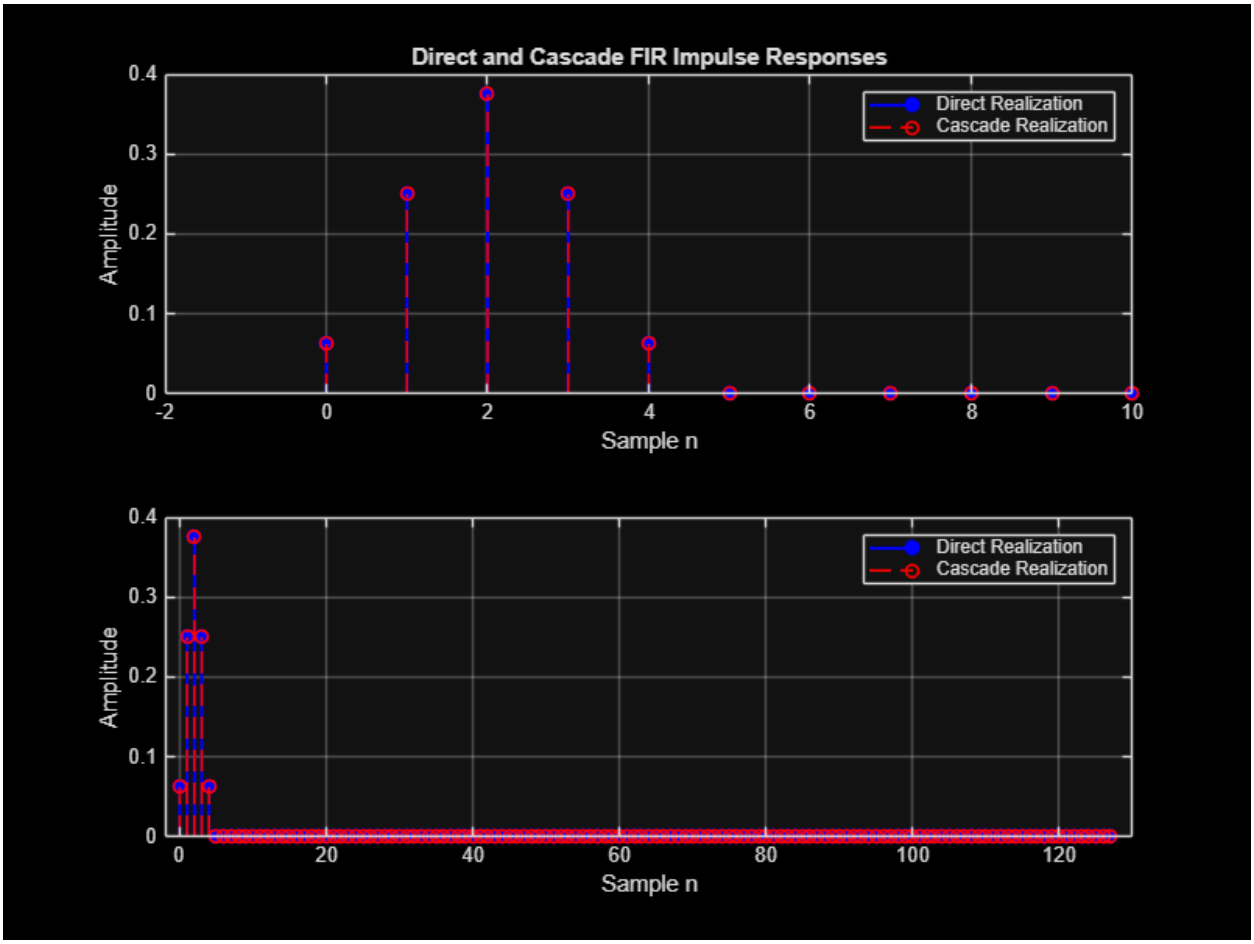
% --- Figure 3: FIR Magnitude Response ---
figure('Name', 'FIR Magnitude Response', 'Position', [150, 150, 700, 400]);
plot(norm_freq, abs(H), 'LineWidth', 1.5);
title('FIR Magnitude Response');
xlabel('Normalized Frequency (\times \pi rad/sample)');
ylabel('|H(e^{j\omega})|');
```

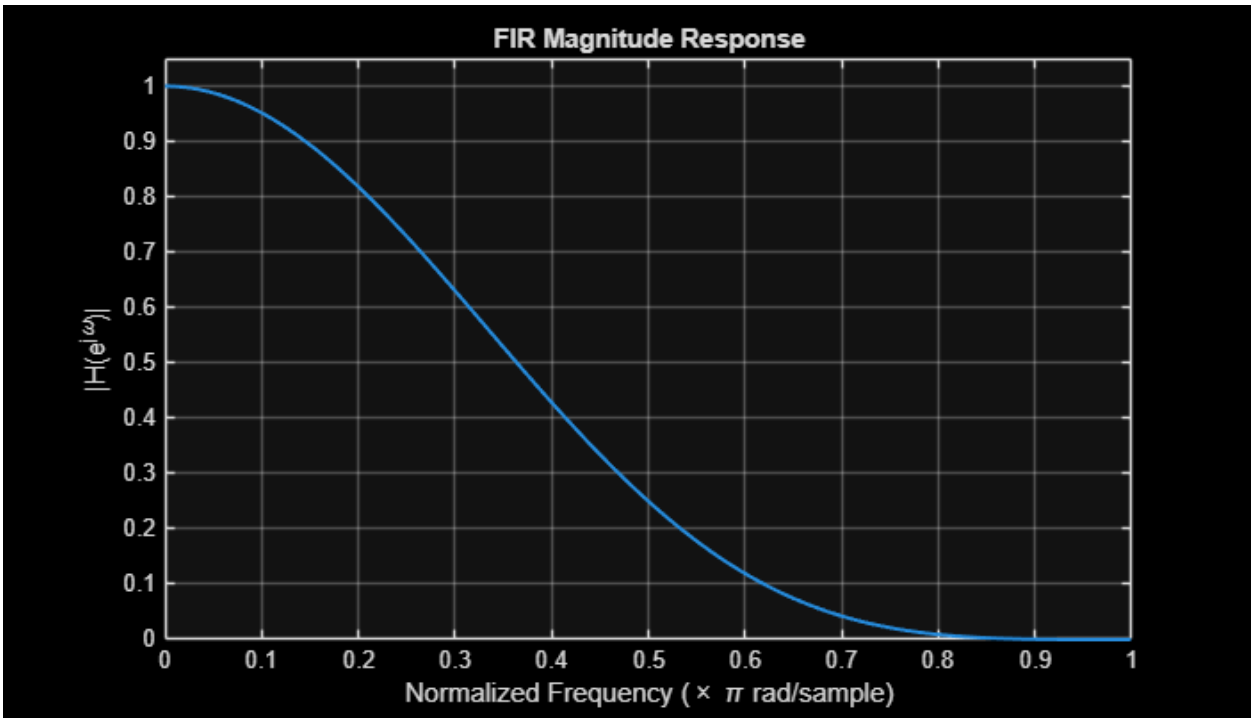
```
grid on;
xlim([0, 1]);
ylim([0, 1.05]);

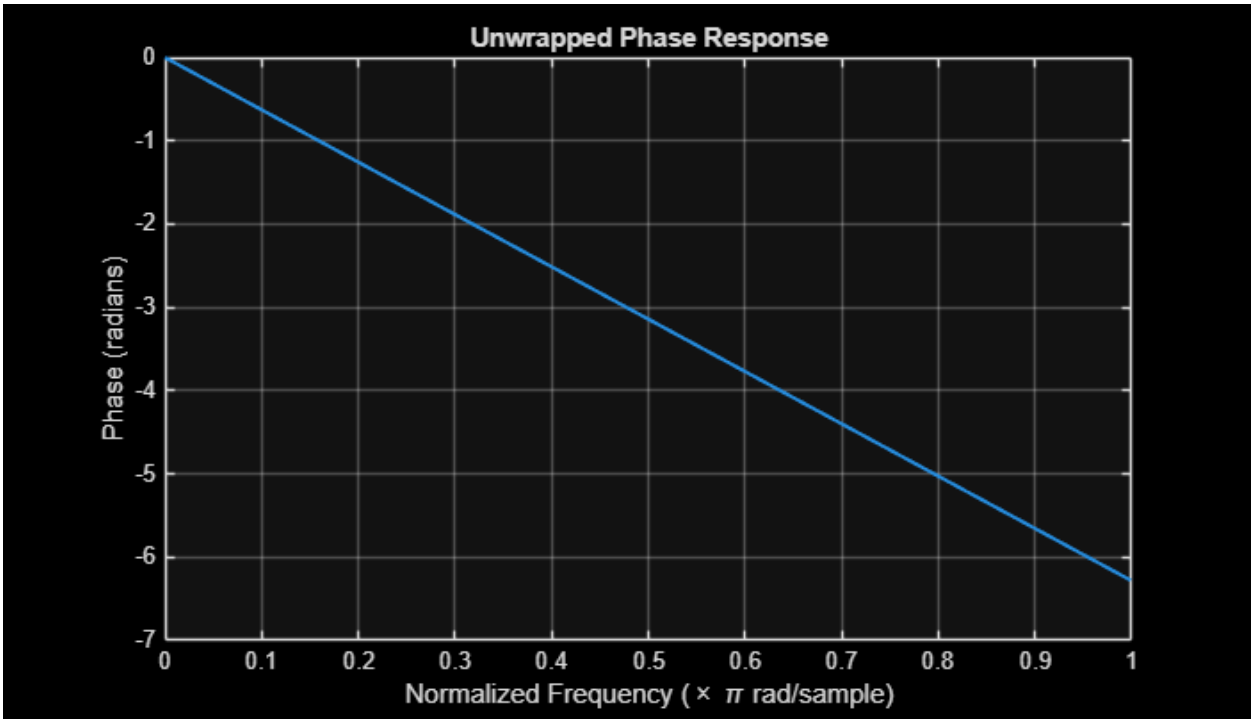
% --- Figure 4: Unwrapped Phase Response ---
figure('Name', 'Unwrapped Phase Response', 'Position', [200, 200, 700, 400]);
plot(norm_freq, unwrap(angle(H)), 'LineWidth', 1.5);
title('Unwrapped Phase Response');
xlabel('Normalized Frequency (\times \pi rad/sample)');
ylabel('Phase (radians)');
grid on;
xlim([0, 1]);

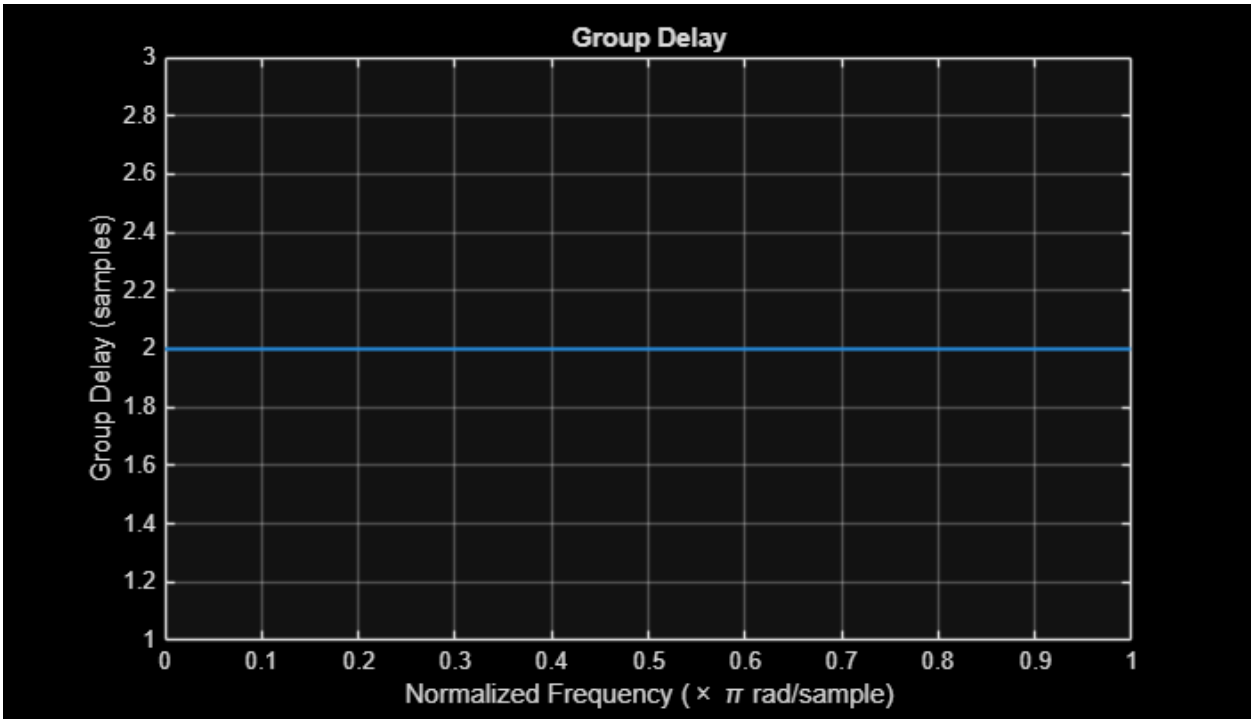
% --- Figure 5: Group Delay ---
[gd, w_gd] = grpdelay(b, a, 1024);
figure('Name', 'Group Delay', 'Position', [250, 250, 700, 400]);
plot(w_gd / pi, gd, 'LineWidth', 1.5);
title('Group Delay');
xlabel('Normalized Frequency (\times \pi rad/sample)');
ylabel('Group Delay (samples)');
grid on;
xlim([0, 1]);
ylim([1, 3]);

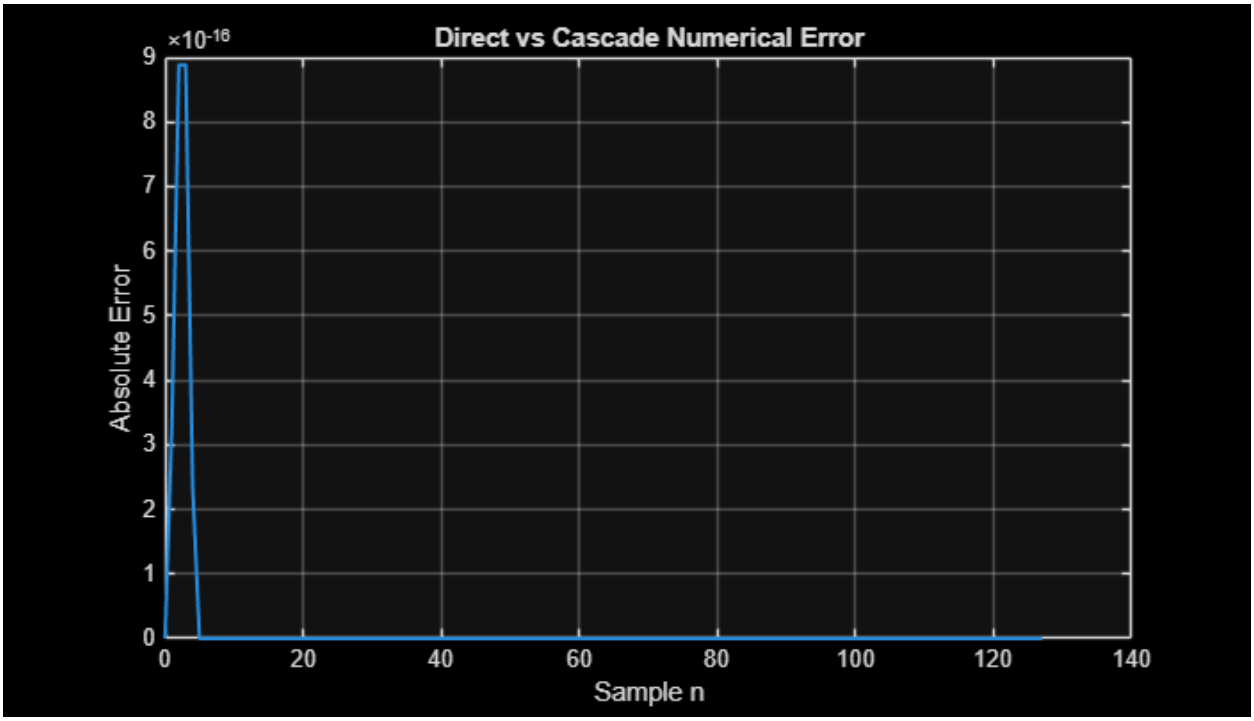
% --- Figure 6: Direct vs Cascade Numerical Error ---
error_val = abs(h_direct - h_cascade);
figure('Name', 'Direct vs Cascade Numerical Error', 'Position', [300, 300,
700, 400]);
plot(n, error_val, 'LineWidth', 1.5);
title('Direct vs Cascade Numerical Error');
xlabel('Sample n');
ylabel('Absolute Error');
grid on;
xlim([0, 140]);
```


Direct and Cascade FIR Impulse Responses
Direct Realization
Cascade Realization
Amplitude
Sample n
0.4
0.3
0.2
0.1
0
-2
0
2
4
6
8
10
Direct Realization
Cascade Realization
Amplitude
Sample n
0
20
40
60
80
100
120


FIR Magnitude Response
|H(e^jω)|
Normalized Frequency (× π rad/sample)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9


Unwrapped Phase Response
Phase (radians)
0
-1
-2
-3
-4
-5
-6
-7
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)


Group Delay
Group Delay (samples)
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)



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